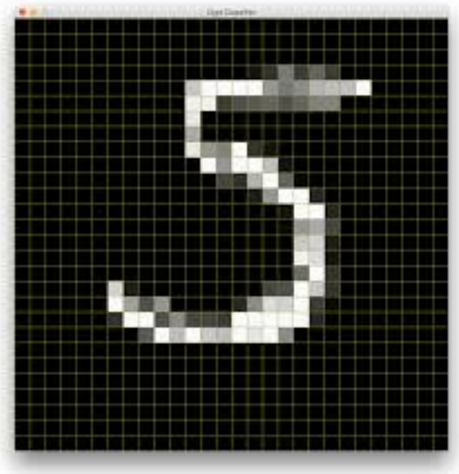
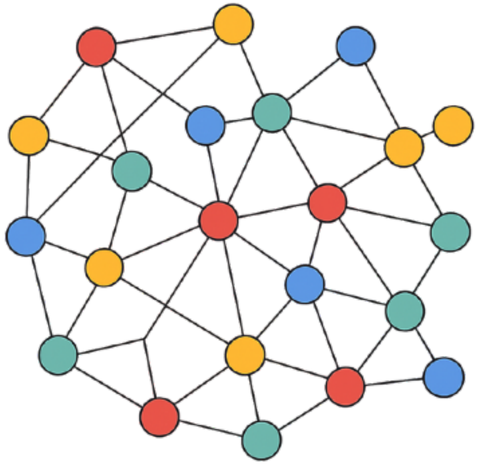
How to organize data?

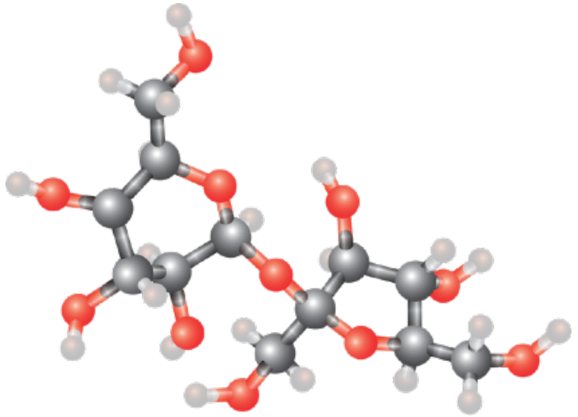
A universal way to think about different modalities of data is in terms of their **symmetries, e.g.**



Images: translations of the pixel grid.



Graph: permutations of the nodes and edges.



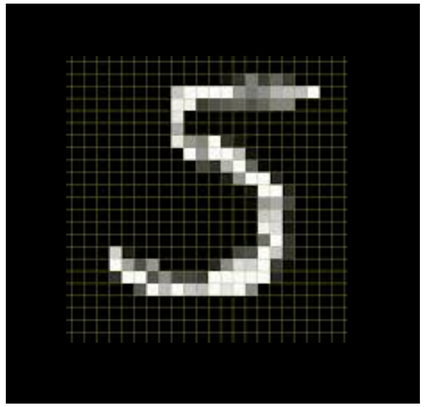
Molecules: translations and rotations of the particle coordinates.

We should use these symmetries to organize our neural networks.

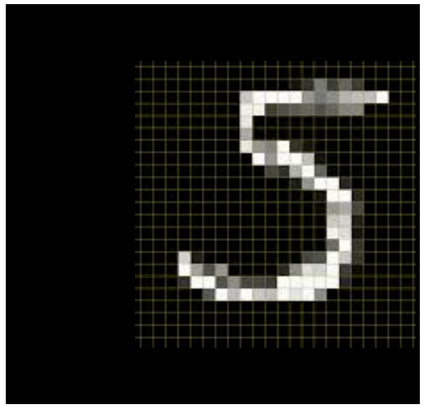
What does it mean to be an image?

Imagine an image as a scalar field $\phi(x)$ on $\mathbb{R}^2$.

$\phi(x) =$



$\phi(x + a) =$



- **Translation**: if $\phi(x)$ is a "5", then so is $\phi(x + a)$.
- **Locality**: the pixels $\phi(x)$ and $\phi(y)$ should be considered "together" by the network if $|x - y|$ is small.
- This idea — that of **translational equivariance** — is encoded in machine learning in a **convolutional neural network,** where layers look like this:

$$W[\phi](x) = \int d^2\Delta \; K(\Delta)\phi(x + \Delta)$$